A framework for studying machine learning methods in healthcare: The First EHR DREAM Challenge

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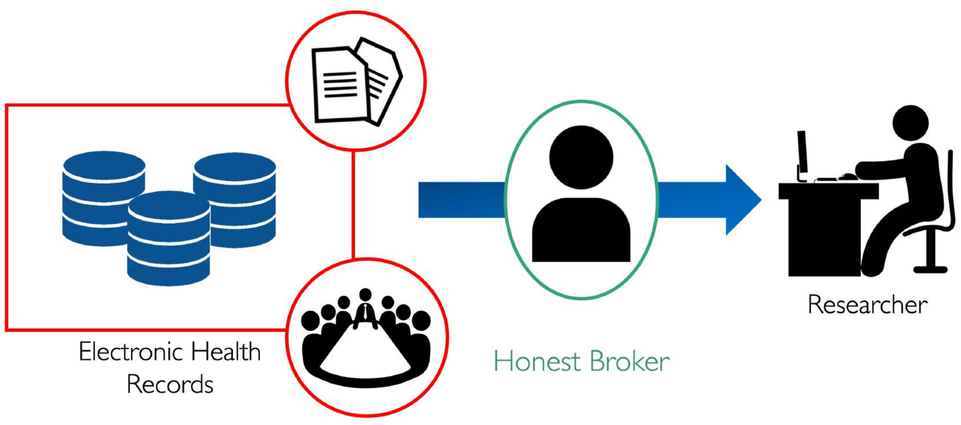
The Model-to-data paradigm enables researchers to train and evaluate models on sensitive health data and allows model assessors to assess bias and accuracy.

An alternative model for sharing sensitive biomedical data

Traditional data sharing paradigm



Due to it's highly sensitive nature, Electronic health record data (EHR) are protected behind firewalls, Institutional Review Boards, Privacy Regulations, and Data Use Agreements. These make using EHRs for building novel predictive models difficult. In addition, when researchers are able to gain access to these data, often through an honest broker who de-identifies the data, re-identifying patients is still a risk even in "de-identified datasets."

Our Solution: The model-to-data paradigm

1. Researchers build models on local environments with synthetic data.
2. Models are containerized (Docker) and submitted to the Synapse collaboration platform.
3. Models are pulled into a cloud environment and validated against synthetic data.
4. Validated models are pulled into the UW secure environment to be trained and evaluated against real clinical records.
5. Predictions are assessed against goldstandard clinical outcomes.
6. Model accuracy metrics are returned to researchers.

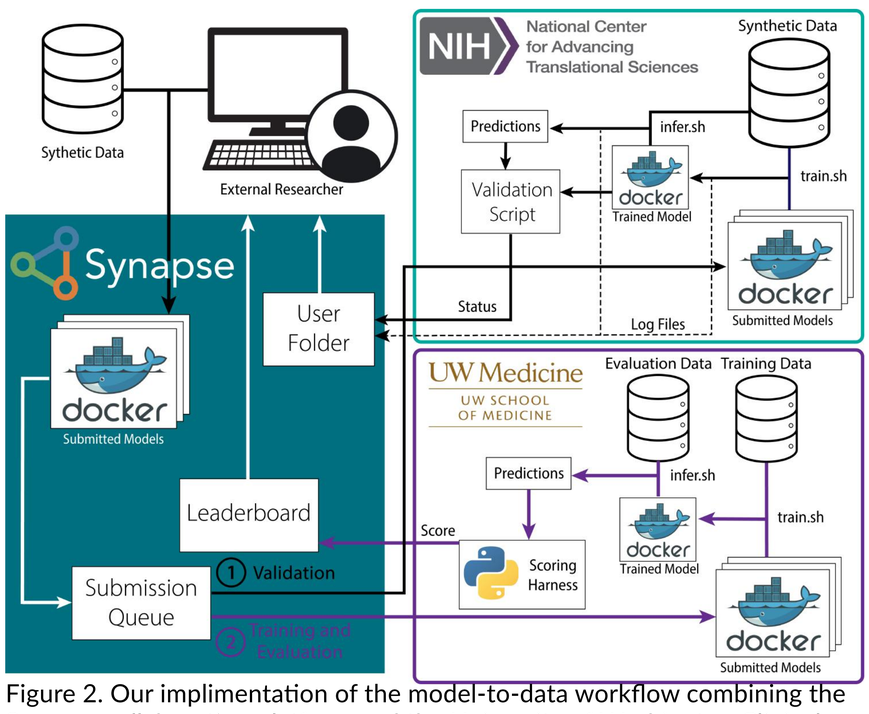


Figure 2. Our implimentation of the model-to-data workflow combining the Synapse collaboration platform and the University of Washington Clinical Data Warehouse.

We wanted to test this novel Model-to-Data infrastructure with a community challenge

EHR DREAM Challenge: Patient Mortality Prediction

Challenge Methods

Participants were asked to address the prediction question: **Given all the past clinical records of a patient, predict the probability that the patient will pass away within 189 days following his/her most recent exam**. All-cause mortality prediction was chosen since it is a well studied prediction question, UW links their EHR records with state death records, and the clinical phenotype is simple to define. 1.3 million patients were hosted within the secure UW environment and made available for training and testing containerized models.

Challenge Results

The challenge was held between September 9, 2019 and February 23, 2020. 25 teams successfully executed 132 mortality prediction models. The highest scoring team (University of Wisconsin biostatistics) achieved an accuracy of 0.947 AUROC and a 0.478 AUPR. (Figure 3) Models had varying accuracies across racial groups, with nearly all models scoring highest on Asian patients. (Figure 4)

Discussion/Conclusion

Challenge participants were able to successfully train and test all-cause mortality prediction models without directly accessing the patient data. Evaluation of clinical prediction models needs to consider stratified demographic evaluation and not simply assess overall model perfromance. Comparing ProActa to PnP_India illustrates how models that have an overall higher score may not have higher performance across stratified sub-populations. Robust evaluation processes are essential to ensure fair and unbiased clinical models.

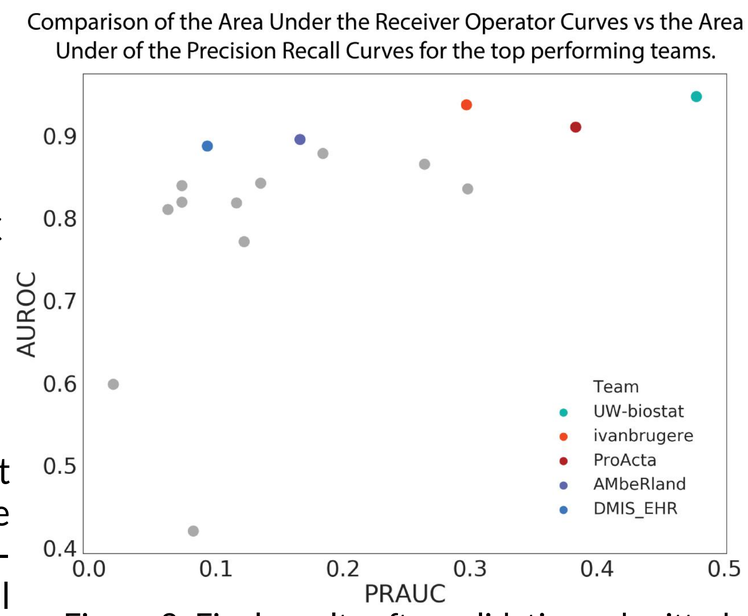


Figure 3. Final results after validating submitted models on a hold out validation set. The top 5 models are colored. Shown are the area under the receiver operator curve vs the area under the precision recall curve.

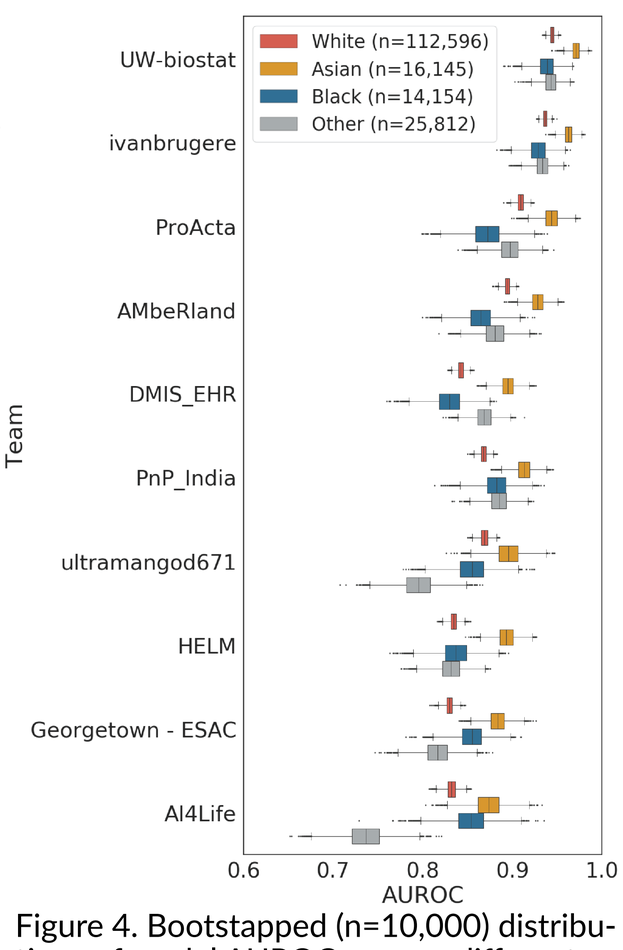


Figure 4. Bootstapped (n=10,000) distributions of model AUROCs across different racial groups for the top 10 teams.

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dreamchallenges.org synapse.org/ehr_dream_challenge_mortality

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